An Analysis of factors that affect a Country's Olympic Performance

BUS-314 Final Project

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1. **Summary of Research Questions and Results**

What is the effect of economic and social indicators on the Olympic performance of the country?
What factors will give a country a higher success rate in the Olympics?
The purpose of this project is to analyze different economic indicators of a country such as population and GDP and social indicators such as being the host country to determine the medal count and the overall success of the country in the Olympic Games. We also looked into the distribution of medals across countries as an indicator of a country's performance in the games and identified patterns in them.

Our analyses found that there is a correlation between Olympic success and larger GDP and population. In most cases, countries with larger GDPs and population sizes tend to have a higher medal count compared to countries with a smaller population and GDP. However, this is not true for all countries.

2. **Motivation and Background**

Studying the relationship between economic and social factors can provide valuable insights into factors that contribute to a country's success in international sports competitions. More specifically, winning in the Olympics is a source of national pride for many countries and success at the Olympics is also viewed as an indicator of the country's overall competitiveness. By studying the relationship between economic and social indicators and Olympic performance, researchers and Olympic committees can gain insight into factors that contribute to a country's success, which can be used to inform policies and strategies aimed at improving an athletes performance in future games. This also helps countries better understand the context in which Olympic success can be achieved.

3. **Datasets**

GDP & Population by Country - from 1960 to 2017
https://exploratory.io/data/kanaugust/GDP-Population-by-Country-from-1960-to-2017-CJb8HUs9Cw

Olympic Data from Winter and Summer Games
https://exploratory.io/data/GMq1Qom5tS/KNH0jzf9HC

To import these datasets you must first open exploratory and create a new project.

- Next, click the plus sign next to "Data Frames" to open a menu.
- Click the option "Data Catalog" that will prompt you with a new window with a search bar.
- To import the first dataset search "GDP & Population by Country - from 1960 to 2017" to find the appropriate dataset.
- The search will prompt you with one result which gives the option to "import" in green on the far right side of the item.
- Click import and a window will open with the columns organized in a table.
- Without clicking anything in the table, click the green import button in the bottom right corner.
- Select a title for the dataset and you have successfully imported the dataset.

The second dataset is very similar with instructions of the first, the only difference is a different search to locate the specified dataset. To find the second dataset search "オリンピックデータ" instead.

4. Ethical Considerations

An ethical consideration we concluded with our data is that some countries or teams may have not qualified or competed in specified years. Some factors that may have led to countries voluntarily missing or being banned from the Olympic Games are doping and boycotts. For instance, Russia was found to be running a state-sponsored doping scheme which resulted in them being banned from competing in the Olympics. Some countries chose to boycott and miss the Olympic Games in certain years due to political conflicts as the Olympic Games have long been a platform for countries to use for their political and economic benefit.

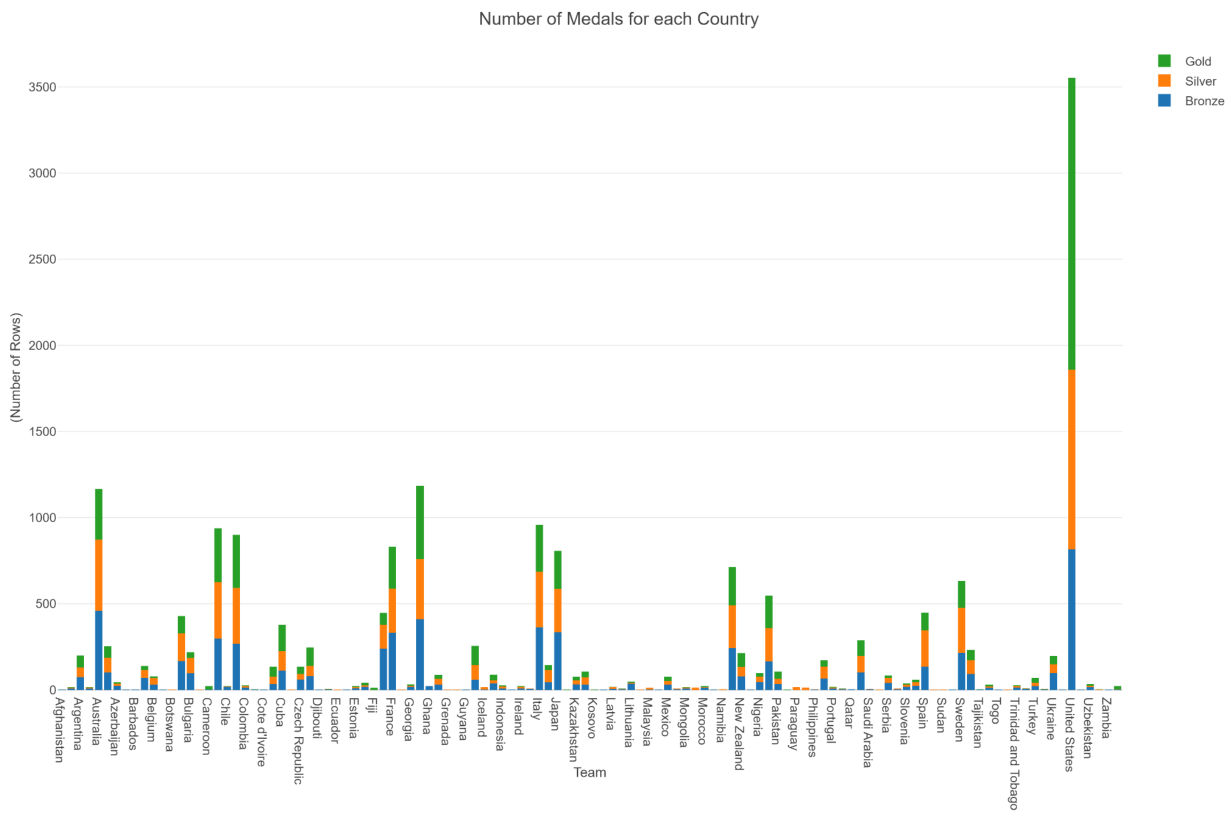
Another ethical consideration concerning the Olympics are the countries' ability to host the Olympic Games. There is bias towards countries that have been able to host the Olympics as it provides their athletes with a home field advantage to perform better – whether that is winning more gold medals or more medals in total. Some factors that may affect the athletes performance in the Olympic Games includes the weather, the presence of a supportive crowd, the qualification standards for athletes, and travel. For home team athletes, these factors may incentivize them to do better than athletes from other countries in the Olympic Games. Additionally, countries that host the Olympics also get a boost in their GDP through the selling of merchandise, tourism, and the traction they gain from social media exposure.

Olympic performance for certain years may also have been affected by the socio-economic factors of a country. For instance, in 1980, the United States led a boycott of the Summer Olympic Games in Moscow to protest the late 1979 Soviet invasion of Afghanistan. In total, 65 nations refused to participate in the games, whereas the rest of the 80 countries sent athletes to compete. Similarly, in 1984, 14 countries led by the Soviet Union boycotted the Los Angeles based Olympic games in retaliation for the US led boycott of the Moscow Games four years earlier. Considering that the Soviet countries did not participate in the Olympics, the U.S easily could have won the medal count in that year.

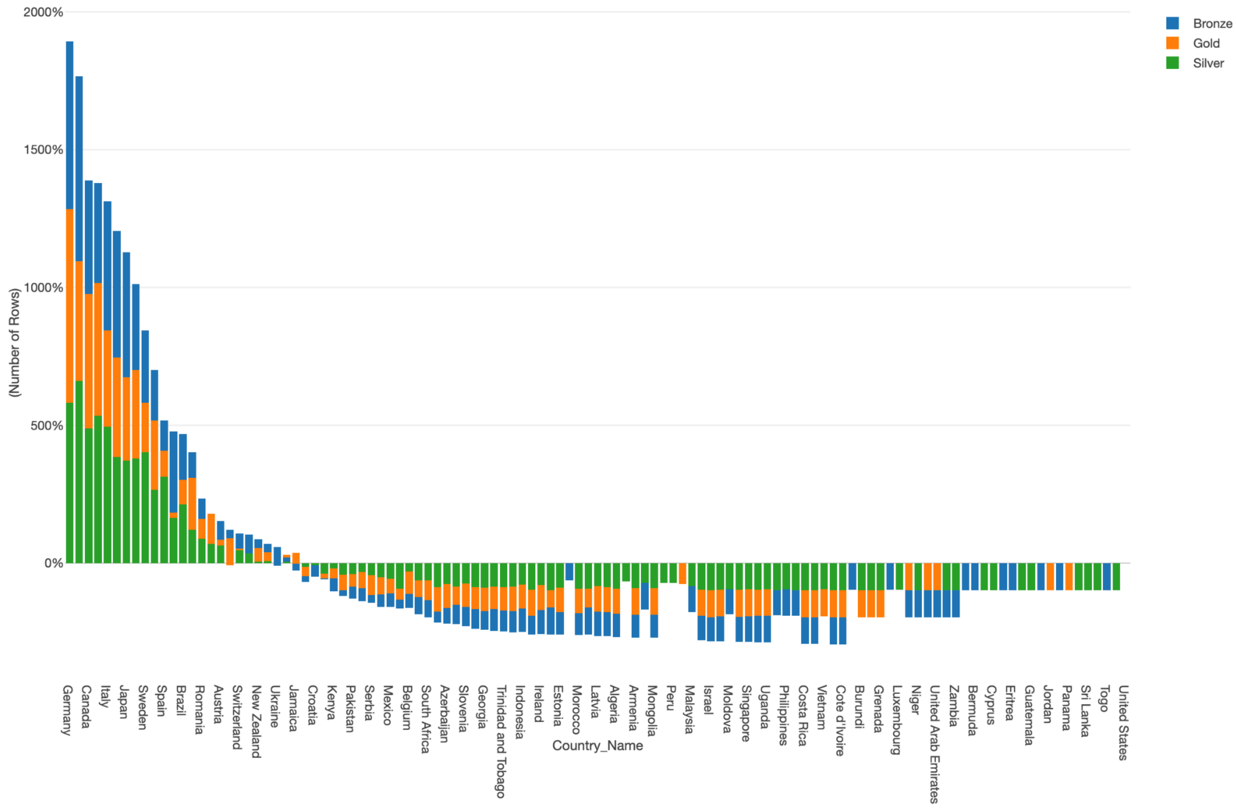
5. Methodology

An important note is that for the GDP and population calculations, we decided to use an average GDP number from the entire timespan of the dataset. This was decided because we only had a few options for this number. We felt that assigning the most recent GDP and population numbers would be irresponsible as it could extremely skew the data for countries that just recently developed or happened to have a recent economic crisis, or have had a large change in population over the time frame of the dataset. Additionally, because the data doesn't date to 2016 for every single country that has won a medal, there could be a problem with the most recent numbers coming from the 1960's for one country while the other is up to date and has a 2016 GDP or population number.

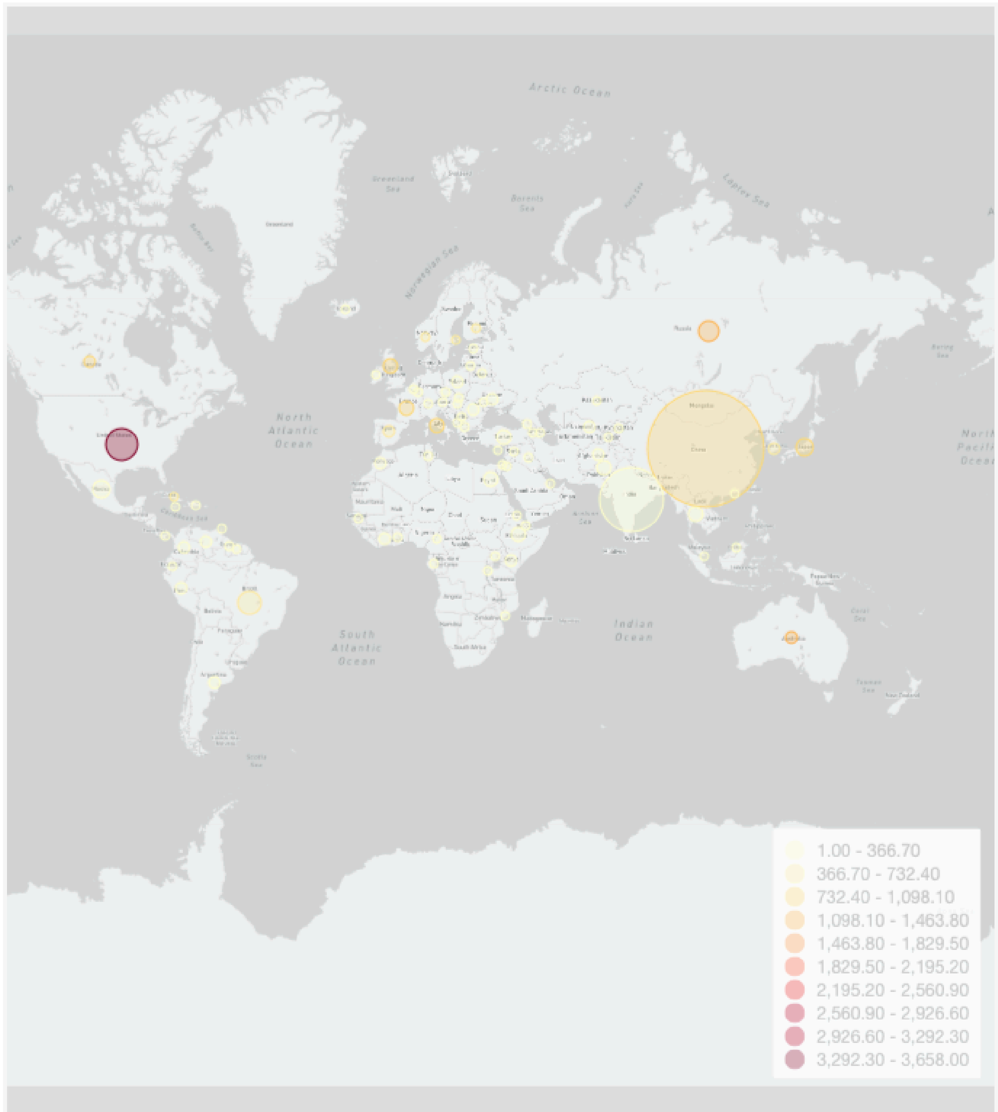
6. Results



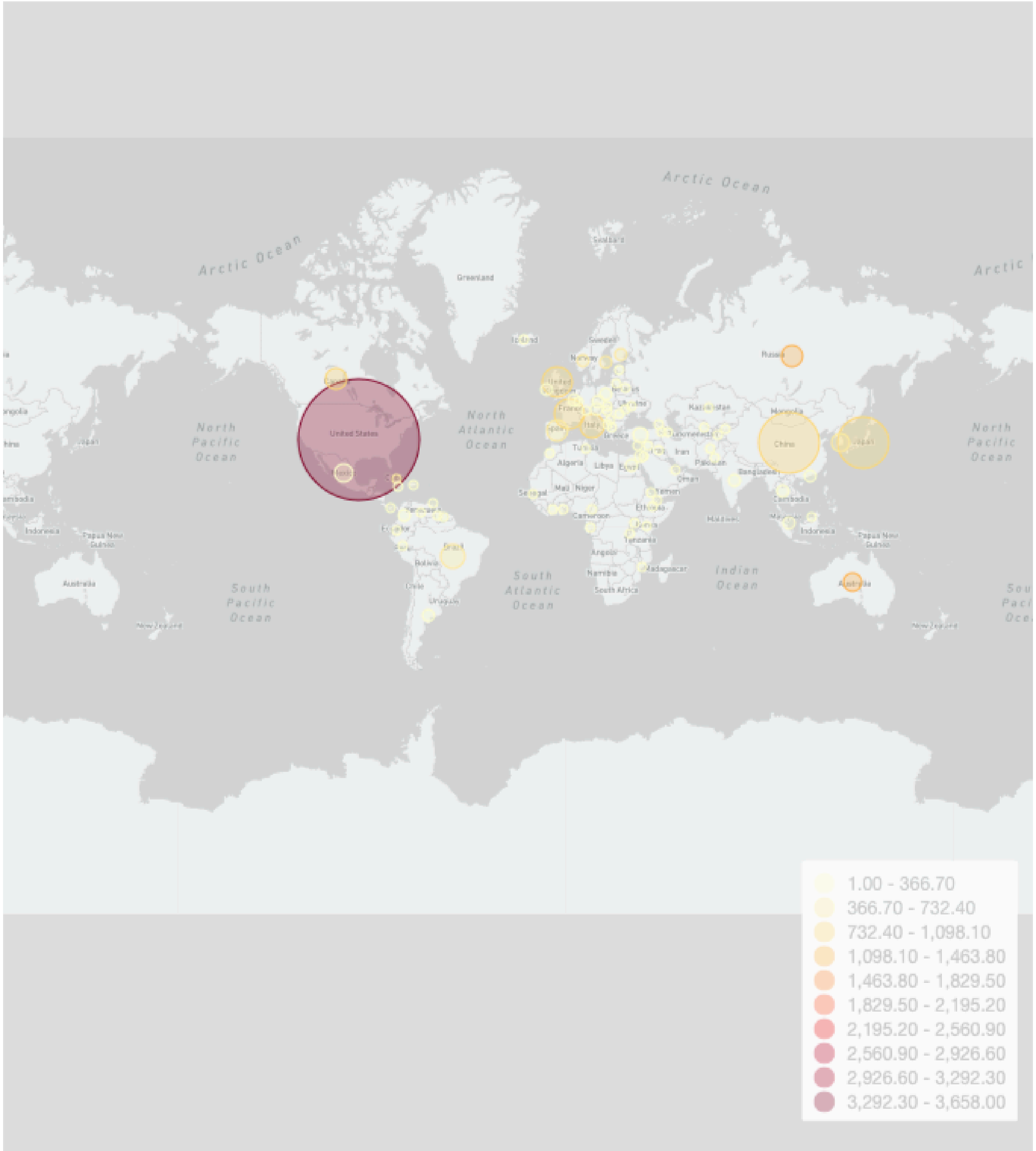
This visualization categorizes the amount of medals in each tier (gold, silver, and bronze) for each country. From this data, we can analyze the distribution each country has between these tiers as well as countries that have achieved more success than others throughout the years. From the data you can see that the United States is far and beyond with the amount of medals won compared to the rest of the world.



This visualization shows the percentage difference from the mean number of medals for each country. Because the United States is far beyond the mean number of medals won compared to the rest of the world, we decided to filter them out to make the data easier to understand. Some interesting observations from this are that it seems that there are a select few countries that are much higher above the mean, which may show us that the data could be a bit skewed by this. Additionally, while many countries are below the mean, there are few that are far lower than the mean. It is also interesting to note that many of the countries above the mean are from Europe. This could be a result of Europe being probably the overall most developed continent in nation by nation terms.



This visualization shows the average population of a country over the time frame of the data with the total number of medals they have won. The average population is shown by the size of the circle in the country, while the number of medals won is based on the color scale from yellow to red. From this data, we can see right away that having a large population appears to have some correlation with winning more medals, but is not always held true. India is an obvious example of this, having one of the largest circles while still having very few medals. However, as seen with China, the U.S., Brazil and Russia, all some of the most populous countries in the world, typically some Olympic success is correlated with having a larger population. This could be due to having a larger pool of athletes to select from as a result of a larger population.



This visualization shows us the average GDP of a country over the time frame of our data, shown by the size of the circle, and the number of medals won by the country, shown by the color based on a scale from yellow to red. As seen above, there seems to be correlation with having higher average GDP and winning more medals. Obvious examples are the U.S., China, Russia, Japan and a few countries in Western Europe like the UK, France and Italy. The implications of this could be that countries with higher GDP have more wealth, and in turn can invest more money into olympic youth training pipelines, better training facilities, better coaches, better equipment, and other factors that can all be important in sport development in a country.

7. Reproducing Results

Data Wrangling

Once both data sets are imported:

- First, in the GDP and Population datasets, in "country_iso3", replace all values "DEU" with "GER". This is to fix a discrepancy between the German name Deutschland and the English name Germany
- Left join the GDP and Population dataset to the Olympics data set, with "NOC" = "Country_iso3c" and "Year" = "year". This combines the two datasets.
- Next, remove columns that may be confusing to have included. For us, this meant removing the two character abbreviation of countries: "Country_iso2c".
- Next, get rid of NAs for certain columns. We removed all NAs for the columns "Medal" because we are only observing countries that have won olympic medals. This allows for "Number of Rows" in any visualizations to be a measure of the number of olympic medals. Next, we removed any NAs for the column "GDP", because these were unnecessary to our analysis.
- Change the data type for the column "Year" to Year instead of numerical.
- Next, we filtered out years that were past 2016, because we did not have any data for any olympics past that date, which means that the GDP and Population data past 2016 were irrelevant to the analysis.
- Finally, Group By "country_name"

Recreating Visualizations

- For the first visualization, create a bar graph. Select X axis as "NOC" (name of country) and Y axis as "Number of Rows". Then, "color group by" using the column "medal".
- To see the % from visualization, first create a bar graph. Make the X axis "country_name" and the Y axis Number of rows. Color (Group By) "medal", and sort by Y axis (descending). Perform a window calculator, where the calculation is the % difference from the mean. Finally, filter out the U.S. because it is such a heavy outlier.
- For the GDP and medals map visualization, create a map. Select World Map- Atlantic Ocean centered, and for the country select "NOC" which is the name of the country. Then, color by the number of rows. Next, change the size of each circle, doing it by GDP. Next, change the color group and sort settings so that there are 10 different groups, and the color palette is yellow-orange-red.
- For the population and medals map visualization, create a map. Select World Map- Atlantic Ocean centered, and for the country select "NOC" which is the name of the country. Then, color by the number of rows. Next, change the size of each circle, doing it by population. Next, change the color group and sort settings so that there are 10 different groups, and the color palette is yellow-orange-red.

8. Collaboration/Bibliography

We did not consult with any outside sources for this project.

9. Reflection

Throughout this project, we learned to further advance our Exploratory skills by searching for our own datasets online and determined a relationship between the two by discovering insights through our research questions. Utilizing the skills we have honed throughout the semester, we constructed various data visualizations to present our insights related to our questions and how there are correlations between different variables in individual datasets. Thus, this project has allowed us to develop our own decision-making and critical-thinking process through the handling of datasets of our own choosing.

Some of the things that would have been helpful to have known before this project are a better understanding of the Olympics and the different factors of the events. Additionally, when considering some of the ethical considerations, having known that some athletes or countries having not competed in certain years would have been useful when analyzing our data. Something that would've benefited us more with the GDP dataset is having a better understanding of the economic picture in each country when considering their performances in the Olympics.

Advice for future students embarking on a data analysis project include conducting a proper and in-depth research prior to pursuing an analysis. Before exhibiting any wrangling or manipulating of your datasets, you should formulate some questions you are trying to analyze and center all the wrangling on the benefit of your question. Another piece of advice is to look through your data and test out different visualizations to find conclusions that are interesting and may apply to your research question. When looking at your visualizations, look for any interesting and unusual points in your data and question them as they often lead to interesting discoveries that you should consider in your analysis.